Microsoft open sources Trill engine for advanced analytics

Microsoft has made one of its internal projects open source. Known as Trill, this project is capable of processing "a trillion events per day," as per company claims.

Trill can be used as a streaming engine, a lightweight in-memory relational engine, and as a progressive query processor.

It was started as a Microsoft Research project back in 2012. Since then, it has been extensively described in research publications such as *VLDB* and the *IEEE Data Engineering Bulletin.*

Till date, Microsoft has used Trill internally for Azure Data products and other mission-critical streaming initiatives throughout the company, including Bing Ads and Halo.

Trill was the first streaming engine to incorporate techniques and algorithms that process events in small batches of data, based on the latency tolerated by the user. It was also the first engine to organise those batches in columnar format, enabling queries to execute much more efficiently than before. To users, working with Trill is the same as working with any .NET library, so there is no need to leave the .NET environment. Users can embed Trill within a variety of distributed processing infrastructures such as Orleans and a streaming version of Microsoft's SCOPE data processing infrastructure.

Trill works equally well over real-time and offline data sets; as a result, it is usually preferred by users who just want one tool for all their analyses.

Trill is also popular among open source developers for the following reasons:

- As a single-node engine library, any .NET application, service, or platform can easily use Trill and start processing queries.
- A temporal query language allows users to express complex queries over real-time and/or offline data sets.
- Trill's high performance across its intended usage scenarios means users get results with great speed and low latency.

By open sourcing Trill, Microsoft has demonstrated that it wants to offer the power of the IStreamable abstraction to all customers the same way that IEnumerable and IObservable are available.

NSA to open source its reverse engineering tool in March

The US National Security Agency (NSA) is planning to open source an internally developed software reverse engineering framework for popular operating systems.

"NSA has developed a software reverse engineering framework known as GHIDRA, which will be demonstrated for the first time at RSAC 2019," states the RSA Conference session description. The conference will be held in March 2019 at San Francisco.

Sources at NSA have added that the tool "…includes all the features expected in high-end commercial tools, with new and expanded functionality that NSA uniquely developed."

GHIDRA dissects binaries for Android, iOS, macOS and Windows, turning them into assembly code that developers can use to help analyse malware and other suspect software. It is built in Java, features a graphical user interface, and runs on Linux, Mac and Windows operating systems.

The existence of the tool came into light in March 2017 when WikiLeaks published allegedly stolen files that revealed the agency was using the tool, *SiliconANGLE* reports.

Those documents show that GHIDRA was first built by the NSA in the early 2000s, and that it has been shared with several other government agencies.

Commenting on NSA's decision to open source GHIDRA, analyst Holger Mueller of Constellation Research Inc. told *SiliconANGLE* that the agency may be hoping that the open source community can help iron out some of GHIDRA's bugs and make it much more reliable.

"The NSA wants to leverage the key benefits of open source, which is more eyes and hands on a set of code," Mueller says.

This is not the first time the NSA is open sourcing its internally developed software. The agency has till now open sourced 32 projects as part of its Technology Transfer Programme (TTP). It has even opened an official GitHub account.

After being demonstrated at the RSA Conference in March, GHIDRA is expected to be released on the agency's code page and GitHub account.